Problem Set 4 Solution

Part One: Computer exercises

1. Download the dataset burger from https://www.openintro.org/data/. The dataset contains responses from 500 individuals about their preferred burger place. The variable `best_burger_place` records each respondent's answer. Suppose we want to investigate whether preferences are equally distributed across the different response categories. For this exercise, drop respondents who answered "Not sure" or "Other" to `best_burger_place`. Use only the remaining response categories in the analysis.

   a. We want to know if respondents are equally likely to choose each burger-place category. What are the null and alternative hypotheses?

   H_0: Respondents are equally likely to choose each burger-place category. Each category has the same population proportion.

   H_A: Respondents are not equally likely to choose each burger-place category. At least one category has a different population proportion.

   b. Make a one-way table showing the number of respondents in each category of `best_burger_place`. Include percentages in the table. Under the null hypothesis, calculate the expected count for each category. Conduct a Chi-square goodness-of-fit test and report the Chi-square statistic, degrees of freedom, and p-value.

```
# data cleaning

burger <- burger %>%
  filter_out(best_burger_place %in% c("Not Sure", "Other"))

# one-way table

count_table <- table(
  burger$best_burger_place,
  dnn = "Best Burger Place"
)
count_table

# output
Best Burger Place
        Fat Burger  Five Guys Burgers
                22                 11
   In-N-Out Burger Tommy's Hamburgers
               343                 54
```

```
      Umami Burger
                 6
```

```
pct_table <- prop.table(count_table) * 100
pct_table

# output
Best Burger Place
        Fat Burger  Five Guys Burgers
          5.045872           2.522936
   In-N-Out Burger Tommy's Hamburgers
         78.669725          12.385321
      Umami Burger
          1.376147
```

```
# expected counts under the null hypothesis

expected_count <- rep(sum(count_table) / length(count_table),
                      length(count_table))

print(expected_count)  # 87.2

# chi-square test

chi_test <- chisq.test(count_table,
                       p = rep(1 / length(count_table),
                               length(count_table)))
print(chi_test)

# output
    Chi-squared test for given
    probabilities

data:  count_table
X-squared = 953.98, df = 4, p-value <
2.2e-16
```

c. Do you reject the null hypothesis at the 5% significance level? Make a conclusion for the hypothesis test in plain English. What do the results suggest about respondents' preferences for burger places?

   Since the p-value is less than the significance level 5%, we reject the null hypothesis that respondents are equally likely to choose each burger-place category.

2. Download the dataset `full_body_scan` from https://www.openintro.org/data/. In November 2014, 1,137 adults were interviewed by telephone about their opinion on the invasive practices airports implemented to uncover potential terrorist attacks. In the file you will find the survey results with responses to the question:

   "Some airports are now using 'full-body' digital x-ray machines to electronically screen passengers in airport security lines. Do you think these new x-ray machines should or should not be used at airports?"

   a. We want to know if the answers significantly vary with a respondent's party affiliation. What are the null and alternative hypotheses?

      H_0: Answers are independent of party affiliation

      H_A: Answers are not independent of party affiliation

   b. Make a two-way table with answers in rows and party affiliations in columns. Include column percentages in the table. Conduct a Chi-square test for independence and report the Chi-square statistic, degrees of freedom, and p-value.

```r
# body <- read.csv("full_body_scan.csv")

# create two-way table

count_table_2 <- table(body$answer, body$party_affiliation,
                       dnn = c("Asnwer", "Party Affiliation"))
print(count_table_2)

# output
                         Party Affiliation
Asnwer                    Democrat Independent Republican
  do not know / no answer       15          22         16
  should                       299         351        264
  should not                    55          77         38
```

```r
pct_table_2 <- prop.table(count_table, margin = 2)
print(pct_table_2)

# output
                         Party Affiliation
Asnwer                      Democrat Independent Republican
  do not know / no answer 0.04065041  0.04888889 0.05031447
  should                  0.81029810  0.78000000 0.83018868
  should not              0.14905149  0.17111111 0.11949686
```

```r
# chi-square test

chi_test_2 <- chisq.test(count_table_2)
print(chi_test_2)

# output
    Pearson's Chi-squared test

data:  count_table_2
X-squared = 4.3576, df = 4, p-value = 0.3598
```

c. Do you reject the null hypothesis? Make a conclusion for the hypothesis test in plain English.

   Since the p-value is higher than 5%, we fail to reject the null hypothesis. That is, we don't have strong evidence that shows an association between answers and party affiliations.

Part Two: Textbook exercises

Question 6.6

a. With a random sample from < 10% of the population, independence is satisfied. The success–failure condition is also satisfied. Hence, the margin of error can be calculated as follows:

$$ME = z^* \sqrt{\frac{p(1-p)}{n}} = 1.96 \times \sqrt{\frac{0.666 \times 0.34}{1018}} \approx 0.029$$

b. A 95% confidence interval for the proportion of adults who think that licensed drivers should be required to re-take their road test once they reach 65 years of age is

$$0.66 \pm 0.03 = (0.63, 0.69)$$

   Since 70% is not contained in the interval, we would reject a null hypothesis where (p=0.7). Therefore, the data do **not** provide evidence that more than 70% of the population think that drivers over the age of 65 should re-take their road test.

Question 6.8

a. The population parameter of interest is the proportion of all Greeks who would rate their lives poorly enough to be considered "suffering", (p). The point estimate for this parameter is the proportion of Greeks in this sample who would rate their lives as such, (p = 0.25).

b. Conditions:

   - **Independence:** The sample is random, and (1000 < 10%) of all Greeks, therefore the life rating of one Greek in this sample is independent of another.
   - **No outliers:** The poll has binary response, so there is no outliers.

- **Success–failure:** (1000 = 250 > 10) and (1000 = 750 > 10).Since the observations are independent and the success–failure condition is met, (p) is expected to be approximately normal.

c. A 95% confidence interval can be calculated as follows:

$$\hat{p} \pm z^* \sqrt{\frac{\hat{p}(1-\hat{p})}{n}} = 0.25 \pm 1.96 \times \sqrt{\frac{0.25 \times 0.75}{1000}} = (0.2231, 0.2769)$$

We are 95% confident that approximately 22% to 28% of Greeks would rate their lives poorly enough to be considered “suffering”.

d. Increasing the confidence level would increase the margin of error, hence **widen** the interval.

e. Increasing the sample size would **decrease** the margin of error, hence make the interval **narrower**.

Question 6.26

a. The hypotheses are

$H_0: \ p_R = p_D$ (Proportions of Republicans and Democrats who support the use of full-body scans are equal)

$H_0: \ p_R = p_D$ (Proportions of Republicans and Democrats who support the use of full-body scans are different)

The pooled proportion can be calculated as follows:

$$\hat{p}_{\text{pool}} = \frac{\text{success}_R + \text{success}_D}{n_R + n_D} = \frac{264 + 299}{318 + 369} \approx 0.82$$

Next we calculate the test statistics,

$$\hat{p}_R = \frac{264}{318} = 0.83 \quad \hat{p}_D = \frac{299}{369} = 0.81$$

$$Z = \frac{\hat{p}_R - \hat{p}_D}{\sqrt{\hat{p}_{\text{pool}}(1-\hat{p}_{\text{pool}})(1/n_R + 1/n_D)}} = \frac{0.83 - 0.81}{\sqrt{0.82 \times 0.18(1/318 + 1/369)}} \approx 0.68$$

and the p-value is

$$P(|\hat{p}_R - \hat{p}_D| > 0.02 | H_0) = P(|Z| > 0.68) = 2 \times 0.2483 = 0.4966$$

Since the p-value is high, we fail to reject H_0. The data do not provide strong evidence that the proportions of Republicans and Democrats who support the use of full-body scans are different.

b. We may have made a Type II error, since we may have incorrectly failed to reject H_0.

Question 6.40

a. Multiply each percent by 701:

	Download	No Download

Position 1	97	128
Position 2	102	130
Position 3	85	159

b. **Prepare.** It is convenient to construct the actual table we care about, which will represent the totals in the three experiment groups:

Group	Total
Position 1	225
Position 2	232
Position 3	244

This now looks like a goodness of fit chi-square problem. As part of our *Prepare* step, we should also construct the hypotheses:

H_0: The chance a user is in any experiment group is equal among the three groups

H_A: The chance of being in one group or another varies by the group

Check. If H_0 were true, then we'd expect 1/3 of the 701 visitors to be in each group, i.e. 233.67 in each group. The 233.67 is greater than 5 for all three groups, satisfying one of the required conditions. The independence condition is also satisfied since each visitor was randomly assigned to a group and only counted once in the table. With the conditions satisfied, we can move forward with conducting a goodness-of-fit chi-square test.

Calculate. We can compute the test statistic and degrees of freedom:

$$\chi^2 = \frac{(255-233.67)^2}{233.67} + \frac{(232-233.67)^2}{233.67} + \frac{(244-233.67)^2}{233.67} = 0.79$$

$$df = 3 - 1 = 2$$

Using software, we can identify the tail area as 0.67.

Conclude. Because the p-value is larger than 0.05, we do not reject H_0. That is, we do not see any evidence that visitor randomization to the groups is imbalanced / malfunctioning.

c. **Prepare.**

H_0: No difference in download rate across the experiment groups

H_A: There is some difference in download rate across the groups

Check. Each visitor was randomly assigned to a group and only counted once in the table, so the observations are independent. The expected counts can also be computed by constructing row and column totals, then multiplying these (and dividing by the table total) to get the expected counts under H_0. Those expected counts are (reading down the first column then down the second): 91.2, 94.0, 98.9, 133.8, 138.0, 145.2. All of these expected counts are at least 5. Therefore, we can use the chi-square test.

Calculate. We can calculate the test statistic as

$$\chi^2 = \frac{(97 - 91.2)^2}{91.2} + ... + \frac{(159 - 145.2)^2}{145.2} = 5.04$$

with $df = (3 - 1) \times (2 - 1) = 2$. This results in a p-value of 0.08, which we can find using software.

Conclude. Because the p-value is less than 0.05, we do not reject H_0. That is, we have not found evidence that there is any difference in the download rate depending on the download link position.

Question 6.42

a. The hypotheses are as follows:

$H_0 : p = 0.5$ (50% of Americans think the Civil War is still relevant)

$H_A : p \neq 0.5$ (A majority of Americans hold an opinion (for or against) that the Civil War is still relevant)

Before calculating the test statistic we should check that the conditions are satisfied:

1. **Independence:** The sample is random, and 1,507 < 10% of all Americans, therefore whether or not one American in the sample thinks the Civil War is still relevant is independent of another.
2. **No outliers**
3. **Success-failure:** 1507 × 0.5 = 753.5 > 10. Since the observations are independent and the success-failure condition is met, is expected to be approximately normal.

The test statistic and the p-value can be calculated as follows:

$$Z = \frac{\hat{p} - p_0}{\sqrt{\frac{p_0(1-p_0)}{n}}} = \frac{0.56 - 0.5}{\sqrt{\frac{0.5 \times 0.5}{1507}}} = \frac{0.06}{0.0129} = 4.65$$

$$\text{p-value} = P(\hat{p} > 0.56 | p = 0.5) = P(|Z| > 4.65) \approx 0$$

Since the p-value is very small, we reject H_0. Since the data show $\hat{p} > 0.5$ and we've rejected H_0, the data provide strong evidence that the majority of the Americans think the Civil War is still relevant.

b. If in fact only 50% of Americans thought the Civil War is still relevant, the probability of obtaining a random sample of 1507 Americans where 56% or more think it is still relevant would be approximately 0.

c. First we need to recheck the success-failure condition using the sample proportion:

$$1507 \times 0.56 = 843.92 > 10$$

and

$$1507 \times 0.44 = 663.08 > 10$$

A 90% confidence interval can be calculated as follows:

$$\hat{p} \pm z^* \sqrt{\frac{\hat{p}(1-\hat{p})}{n}} = 0.56 \pm 1.65 \sqrt{\frac{0.56 \times 0.44}{1507}} = (0.54, 0.58)$$

We are 90% confident that 54% to 58% of all Americans think that the Civil War is still relevant. This agrees with the conclusion of the earlier hypothesis test since the interval lies above 50%.

Question 6.50

a. Chi-squared test of independence.

b. The hypotheses are

H_0: Caffeinated coffee consumption and depression in women are independent

H_A: Caffeinated coffee consumption and depression in women are dependent

c. Depression: $\frac{2607}{50739} = 0.0514$

No depression: 1 - 0.0514 = 0.9486

d. We have

$$E = \frac{2607 \times 6617}{50739} = 339.99 \approx 340$$

$$\frac{(O-E)^2}{E} = \frac{(373-340)^2}{340} = 3.20$$

e. $df = (R-1) \times (C-1) = 1 \times 4 = 4$, and p-value < 0.001

f. The p-value is small and we reject H_0. The data provide convincing evidence to suggest that caffeinated coffee consumption and depression in women are associated.

g. Yes, this is an observational study. Based on this study we can't deduce that drinking more coffee leads to less depression. There may be other factors, lurking variables, that cause decreased depression in women who drink more coffee.